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When Battery-Free Interfaces Support Design Changes Information, Qualified Observations and Event-Time Service

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Abstract

Battery-free design decisions depend on when energy becomes usable, which state survives a reset and what a receiver must learn. A representation that explains one successful trajectory can omit the mechanism activated by the next legal design change. We formalize this distinction as current-verdict versus intervention sufficiency, give an elementary factorization condition, and exhibit ideal storage-clamp and reset-memory counterexamples. We then implement a bounded route from acquired source data through independently qualified, typed observations to actual design consumers. A matched comparison uses four fixed digital tasks and three formal repetitions per condition. Average-energy inputs yield 8/12 complete-service passes; rich-static, complete-input, executable-response and equal-information table inputs each yield 12/12. Prospective live observation yields seven completed chains and five incomplete groups, retaining all twelve in the denominator. Six completions select supplied fields and one constructs a relation; none supplies a physical measurement. A secondary event-service study separates event matching from timestamp accuracy: two one-second designs match all eight events and pass energy yet fail encoded-time bounds at 1.004 and 1.103 s. Three 0.75-second designs pass; four other episodes terminate without a submission. These results identify information and admission boundaries without demonstrating superiority over complete static information. Physical qualification remains UNKNOWN, and prior product failures remain unaccepted.

1 Introduction

A battery-free device must turn an intermittent supply into a particular service: capture an external event, preserve the relevant state, and deliver the required information at the required time. Having enough energy in total does not specify whether a rail becomes ready before a deadline, whether a transmit pulse causes a reset, or whether the event is still pending when the device recovers. The interface presented to a designer determines which of these relations can be evaluated before a legal change is made.

The distinction matters even when the current service succeeds. Two worlds can produce the same present voltage and event traces while having different unexcited storage clamps. Reducing capacitance activates those clamps and separates their outcomes. Two pending-event memories can likewise behave identically until a reset is introduced. These examples do not imply that static descriptions are intrinsically inadequate. A static description of the relevant laws, parameters, initial state and legal intervention domain can already answer the prospective question [4, 5].

An observation interface offers another route: acquire a source, select or construct a relation, qualify its domain and units, and feed the result into a design consumer. That route has an operational burden of its

own. A useful equation with an invalid path can fail admission; a correctly computed Boolean can fail to match a numeric consumer branch. A qualified observation can also leave the final design unchanged, or support a design that complete static information already permits. Source acquisition, typed consumption, service success and incremental benefit therefore require distinct evidence [6, 7, 12].

We study these distinctions with sealed, executable digital BF cases. The first part states the information requirement and instantiates two structural counterexamples. The second implements the acquisition-to-consumer route and compares it against average-energy, rich-static, complete-input and executable interfaces under matched budgets. The main formal comparison has four fixed tasks and three repetitions per condition. Its strongest negative result is central: rich-static information already obtains 12/12 complete-service passes, whereas live observation completes seven chains and leaves five groups incomplete. The equal-information serialization comparison gives the same 12/12 service and 65 requests in both arms [11, 12].

A secondary vibration-event study tests a different service boundary. Correct event matching and energy accounting do not ensure correct encoded event times. Two designs fail one-second timestamp tolerances even though their messages arrive promptly. We retain these submitted failures separately from four token-censored episodes with no design submission [18].

The contribution is a connected account of what information must distinguish, how an observation must reach a real consumer, and where successful digital checks stop supporting the claim. It includes the theoretical condition, mechanism witnesses, strong static controls, actual failed and completed observation chains, and portable saved-data reconstruction. It supplies no claim of priority over general sufficient-statistic or abstraction theory, no advantage over the successful complete static controls, and no hardware qualification. The study concerns public development apparatus rather than an official benchmark or a population of unseen physical devices.

2 Related Work

Intermittent systems research addresses the consistency of execution across power failures. Alpaca organizes a program into tasks, privatizes shared data and commits task effects so interrupted execution can restart without exposing partial updates [1]. This establishes a concrete recovery mechanism; it does not make a current energy summary sufficient for a different capacitor or establish that an event reached its receiver. Our retained-state witnesses use an idealized latch and do not implement or evaluate Alpaca.

Curricle specifies intermittent idempotence requirements through a type system and information-flow qualifiers, checking programs against those requirements [2]. Its program-level consistency question complements our admission question: whether an observation has declared units and domain, is independently qualified on the supplied source, and reaches a compatible design consumer. Passing these interface checks does not prove intermittent idempotence, arbitrary program correctness or physical validity.

Counterexample-guided abstraction refinement (CEGAR) refines an abstraction when an abstract counterexample is spurious in the concrete system [3]. Our formulation shares the use of distinguishability to explain missing information. Its role is narrower: two legal digital worlds with the same representation require different verdicts after a specified intervention. Proposition 1 is an elementary factorization condition, not a new general theory of abstraction or a CEGAR algorithm. We supply neither a general model checker nor an automated refinement guarantee. The empirical contribution is the source-bound comparison among complete static information, executable feedback and qualified live observations, including failed chains and receiver-side service conditions.

3 The information required by a design consumer

3.1 Current verdicts and legal interventions

Let T fix the external service and its evaluator, $w \in \mathcal{W}$ denote a world including device laws and state rules, and d_0 be a current design. The complete nominal verdict is $V_T(w, d) \in \{\text{PASS}, \text{FAIL}\}$. The representation $R_{d_0}(w)$ is the information legally available at the current design. It may contain parameters, records, source

definitions or traces; it need not be a scalar. A legal intervention δ produces $\delta(d_0)$. Comparisons below use the same T and only interventions jointly legal in the compared worlds. A change in the success criterion is not a design intervention.

Current-verdict sufficiency requires

$$R_{d_0}(w) = R_{d_0}(w') \implies V_T(w, d_0) = V_T(w', d_0). \tag{1}$$

Intervention sufficiency on a declared common domain $\mathcal{D}$ requires the stronger implication

$$R_{d_0}(w) = R_{d_0}(w') \implies V_T(w, \delta(d_0)) = V_T(w', \delta(d_0)) \quad \text{for every } \delta \in \mathcal{D}. \tag{2}$$

Both are properties of a representation, consumer, world domain and task together. A field set sufficient for energy balance need not be sufficient for event retention or timestamp accuracy. Nor does either property promise that a finite-budget language-model procedure will use the information well.

Proposition 1 (Elementary factorization condition)**.** *On a fixed common legal domain, an exact prospective predictor $g(R_{d_0}(w), \delta) = V_T(w, \delta(d_0))$ exists if and only if the verdict for each δ is constant on every equivalence class of worlds sharing R_{d_0}.*

Proof. If g exists, identical representations and the same intervention give it identical arguments, so their required verdicts must agree. Conversely, if the verdict is constant on each class, define g by that class's verdict for the requested intervention. Constancy makes the definition independent of the representative world. This is an information condition; the proof does not establish an efficient implementation or calibrated world model. □

An interface can return UNKNOWN when its equivalence class contains both verdicts. Abstention then differs from an incorrect PASS or FAIL. We also distinguish this epistemic answer from INCOMPLETE, the operational outcome when an episode supplies no admitted final execution. Physical qualification UNKNOWN is a third boundary: it describes absent calibration evidence even when the digital verdict is well defined.

3.2 Refinement and the strong static alternative

Acquiring an observation O refines the representation to (R, O). It is informationally useful for a particular intervention only if it separates worlds whose prospective verdicts were previously ambiguous, or supplies another needed input to the consumer. Provenance, domain and unit checks establish admissibility of the value. They do not, by themselves, establish that the value distinguishes the relevant worlds or changes service.

A complete static model is an important limiting case. If its supplied parameters, laws and initial state determine the execution for every legal d, then executing or reasoning about that model defines an exact g on the stated ideal domain. A dynamic response can make a computation easier to inspect, and an observation can reduce the information a consumer must handle. Neither has an automatic informational advantage over that complete static model. A finite-budget comparison must measure any computational or operational advantage rather than infer it from the word "executable."

This separates three interface families evaluated here. A current trace describes one realized behavior. A declarative static model additionally specifies how permitted changes affect that behavior. An executable response computes candidate-specific consequences. A qualified observation is a source-bound intermediate fact delivered through a typed edge. These are distinct interfaces, but their useful information can overlap completely. The static-grid checks and matched rich-static arm explicitly test that overlap [4, 12].

3.3 Service has several independent predicates

The BF service consumer combines energy feasibility, event capture, retained state, message validity and timing. An ordered retention witness requires the same event to be captured before a reset and first validly received afterward. A reset after delivery does not satisfy that witness. Similarly, matching an event identifier

does not establish a correct encoded event time. For a confirmed event at t^*, receipt at t_{rx} and encoded stamp $\hat{t}$, an O-style service includes both

$$0 \le t_{\mathrm{rx}} - t^* \le L, \qquad |\hat{t} - t^*| \le \epsilon, \tag{3}$$

in addition to energy and state predicates. The later experiment uses $L = 10$ s and $\epsilon = 1$ s. The receive clock and the encoded stamp are different quantities. Passing the first inequality cannot replace checking the second [17, 18].

4 Mechanism counterexamples and their sufficient conditions

4.1 An unexcited storage clamp: E2

Two ideal worlds receive 20 μW for 2 s and face a 30 μJ service transaction at 2.5 s. Both start with zero stored energy, use $C = 10$ μF and must retain the energy associated with a 1 V minimum rail. Their clamps are 3 V and 5 V. The current maximum voltage is $\sqrt{8} = 2.828$ V, so neither clamp activates. Their complete current observed event and rail traces coincide and both services pass.

The common legal change sets $C = 4$ μF within the declared $[4, 12]$ μF domain. At 3 V, only 18 μJ can be stored; 22 μJ spills and only 16 μJ is available above the 1 V reserve. Service fails with a -14 μJ margin. At 5 V, all 40 μJ is retained and the margin is 8 μJ, so service passes. The task, source, requested transaction and common change are identical. Thus the current trace does not distinguish a mechanism needed for the prospective verdict [4].

An enriched static description resolves the ambiguity. Under the exact assumptions that all input precedes service, the capacitor is ideal and there is no leakage or intervening workload, define

$$M(C) = \min\left(E_0 + P_{\mathrm{in}} t_{\mathrm{in}}, \frac{C V_{\max}^2}{2}\right) - \frac{C V_{\min}^2}{2} - E_{\mathrm{service}}. \tag{4}$$

Service passes exactly when $M(C) \ge 0$. The first term is the stored energy after the source ends; the next is the required residual energy. With no other transaction between those times, their difference is exactly the budget available to service. The sufficient object is the functional $M(\cdot)$ and its parameters on the legal domain, not merely its value at the current C. Both worlds have the same current margin. A current scalar margin alone therefore retains the counterexample.

4.2 An unexcited reset transition: E3

The second pair captures event 1 at 1 s and attempts its single transmission at 4 s, before a 5 s deadline. A volatile pending memory and a retained pending memory both deliver in the current uninterrupted execution. The same legal change inserts a power outage from 2 to 3 s. Volatile memory clears the pending event and delivers none; retained memory delivers one. External event timing and the transmission schedule remain fixed. This witness changes a state transition rather than scaling an energy total [4].

The sufficient static state description must state capture, reset and send semantics and their order. If pending state survives each intervening reset, and the scheduled send can execute after restoration, the captured event remains available. If the only pending copy is cleared and no recapture is possible before sending, it does not. These are conditions of the ideal machine, not a claim that an arbitrary nonvolatile device has free writes, atomic persistence or a qualified energy budget.

The saved analytic and numerical clamp implementations and the separate memory interpreter agree on eight current/intervention cases at both 1 ms and 0.5 ms settings: sixteen resolution-specific comparisons. Maximum reported energy discrepancy is 2.84×10^{-12} μJ. The enriched static model agrees on all 28 exposed grid points, and an additive static control agrees on five service costs spanning PASS and FAIL. Removing the unexcited clamp or reset-retention fact restores aliasing in these encodings. These checks establish software consistency under common ideal assumptions, not independent physical truth or universal necessity of a particular field name [4].

Table 1: E2–E3 structural witnesses: both worlds pass before a common legal intervention, then separate. These are two exposed ideal pairs, not statistical repetitions.

Mechanism	World	Current	Changed	Changed events
Clamp	low-clamp	PASS	FAIL	0
Clamp	high-clamp	PASS	PASS	1
Reset memory	volatile	PASS	FAIL	0
Reset memory	retained	PASS	PASS	1

4.3 Connecting the witnesses to transient BF behavior

Seven separate saved BF controls exercise source timing, startup, loaded rail, reset and reception (Table 2). Their transient model uses ready/off thresholds, boot energy, finite TX current and duration, and the loaded-rail relation

$$V_{\text{load}} = \sqrt{2E/C} - I_{\text{TX}} R_{\text{ESR}}. \tag{5}$$

A failed energy or rail gate can reset the program and clear volatile pending state. The causal path is load to stored energy and loaded rail, then reset and program behavior. The source is prescribed, so no load-dependent source feedback is established [5].

Table 2: Seven saved digital BF reference controls. First reception after 100 ms fails the deadline; a passing rail gate without a received event is insufficient. Physical qualification is UNKNOWN throughout.

Control	Input (μJ)	RX (ms)	Service	Rail
Fast pulse	150	15.8	PASS	PASS
Slow pulse	150	54.8	PASS	PASS
High ESR	150	–	FAIL	FAIL
No energy	0	–	FAIL	PASS
Late pulse	150	105.8	FAIL	PASS
Latched reset	210	46.9	PASS	PASS
Volatile reset	210	–	FAIL	PASS

Fast, slow and late pulses each supply 150 μJ, yet their first receptions differ; the late one misses the 100 ms deadline. A 150 Ω ESR control prevents reception despite the same source integral. The no-energy control passes its rail check vacuously and fails service. The reset pair supplies 210 μJ with the same 40 μJ shunt and postponed transmission, but pending-event survival differs. Its retained mechanism is an explicitly ideal passive mechanical latch able to capture without stored electrical energy. It is not a model of a qualified commercial contact device or a powered NVM write.

All seven agree with their predeclared control expectations, including the intended failures. That agreement is different from seven service passes. The full input laws and source calibration limits accompany the saved data; physical qualification remains UNKNOWN for every control. These transient examples connect the structural issue to BF operation without turning the two ideal witness models into a complete physical node.

5 From a missing relation to an executed consumer

5.1 A bounded observation language and external qualification

The observation interface makes the information path explicit (Figure 1). The participant can list and acquire permitted sources, propose either a selected field or a constructed relation, and submit a typed design policy. A proposal states the source identifier, JSON Pointer paths, units, applicability domain and intended relation. Selecting a supplied field and constructing a new expression are recorded as different operations. Neither label alone establishes a new physical fact or a novel scientific relation [6, 9].

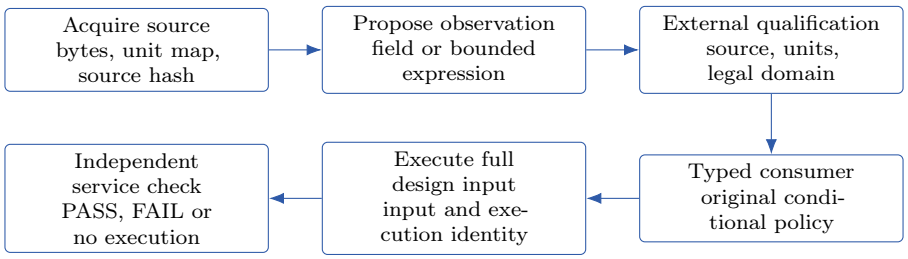


Figure 1: The observation-to-consumer chain. Rejection can stop the chain before any final service execution. Every attempted action and its cost remain recorded. A source-bound value or a consumed observation alone is not service or benefit evidence.

Construction uses a bounded JSON abstract syntax tree rather than arbitrary Python. The interpreter admits at most 64 nodes, depth eight, an 8 MB JSON source and 32,768 vector elements. A construction must contain a source field and an allowed operator. Unit dimensions must agree across operations; the declared output unit must match the root, and declared numeric domain constraints are checked against source values. Missing fields, unsupported domains and invalid arithmetic lead to rejection or UNKNOWN. These limits bound interpretation; they do not calibrate the supplied source.

Authority is outside the proposer. The host binds the original source bytes, proposal code and input, provider/run identity and actual accounting. An independent interpreter checks the source relation. A proposer cannot qualify its own output by supplying an authority record or a success label. The consumer receives only an admitted observation and the original policy, then chooses among at most three strictly typed equality branches. Boolean true and a numeric duration are different values. The complete selected input, observation edge and actual execution identity are preserved before the separate service verdict is read [6, 8, 9].

This design distinguishes acquisition from consumption. Echoing a value in a report is insufficient. A valid consumption receipt shows how that value entered the executed design. It still does not establish causal necessity: the same design might be chosen with complete static inputs, and an observation can be consumed without changing the design. A claimed benefit would require a separately frozen comparison of alternatives with the same task and allowance.

5.2 A concrete constructed relation and its actual consequence

An earlier actual E5 attempt acquired component, excitation and saved-activity records, then constructed an energy comparison. Its first proposal omitted the leading slash required by JSON Pointer. The original host counted that invalid proposal against the one-proposal limit; the model's already generated correction was then rejected at the cap. The live attempt remains rejected: no qualified observation, no real consumer execution, and no proposed-design scientific execution. Its six SETTLED requests and USD 0.0644115 upper-bound rate estimate remain charged [7].

A separately authorized record labeled `POSTHOC_HOST_ADAPTER_REEVALUATION` later evaluates that existing corrected call. It makes no new model request and does not replace the failed live attempt. The model's retained relation is

$$E_{\mathrm{TX}} = I_{\mathrm{TX}} \Delta t_{\mathrm{TX}} V_{\mathrm{out}} = 28.8\ \mu\mathrm{J}, \tag{6}$$

$$E_{\mathrm{reserve}} = \tfrac{1}{2} C (V_{\mathrm{ready}}^2 - V_{\mathrm{off}}^2) = 23.0\ \mu\mathrm{J}, \tag{7}$$

$$b = [E_{\mathrm{TX}} > E_{\mathrm{reserve}}] = \mathrm{true}. \tag{8}$$

The supplied record uses 8 mA, 2 ms, 1.8 V output, 10 μF capacitance and 2.8/1.8 V ready/off thresholds. The source-bound domain allows 4–22 μF and 0–45 ms earliest TX, with TX current fixed at the record's value. A separate Decimal interpreter agrees on the Boolean and verifies source dependence over legal capacitance

values. The qualification criterion was revised post hoc from a paired-world-specific test to dependence across legal design values. That intervention and its scope remain explicit [8].

Before an admitted observation, the consumer keeps UNKNOWN and the original 10 μF, 45 ms design. Qualified Boolean true selects the model's original 15 μF, 45 ms branch. The actual full service changes from FAIL with no received frame to PASS with first reception at 46.9 ms. Both energy ledgers pass, so the energy flag alone cannot explain service. The selected design avoids a pre-deadline brownout; it does not demonstrate retention of a pending event across one. A complete static-information design already passes at 10 μF and 0 ms, receiving at 5.6 ms. The unused false branch's service is not established by these receipts [8].

The relation is therefore a concrete derived fact that reaches a real design consumer and has an executed engineering consequence. It is not prospective live-E5 success, a physical measurement, proof that capacitance must increase, or an advantage over complete static information. Its energy inequality omits startup work, losses and event order and cannot serve as a complete service evaluator. Later adapter-v2 regression records reject malformed syntax before charging the valid-proposal counter, while bounding syntax rejections and retaining a one-valid-proposal limit. Those new-version checks preserve the original failure [9].

6 Matched digital experiment

6.1 Tasks, outcome and strong controls

The main comparison uses four fixed tasks. Pulse-fast is a static-sufficient control. Volatile-reset and latched-reset use the transient BF apparatus with different pending-state semantics. Their legal design variables are capacitance and earliest permitted transmission, under frozen task-specific ranges. Complete service uses independent energy and receiver checks plus the registered ordered-event analysis. PASS denotes admitted complete service, FAIL an admitted final design that fails it, and INCOMPLETE no admitted final execution [10, 11].

The fourth task is a preregistered C-lite single-query derivative. It supplies 20 μW ideal DC for 30 s, 1 μW idle load, zero initial energy and a 50 μJ reply. A query at 2 s must receive its matching reply by 7 s with maximum data age 5 s. Capacitance is 25–100 μF at an ideal 2 V ceiling; ESR, boot energy and boot delay are zero and harvest efficiency is one. Gates require query issuance, capture, matching token, causal order, deadline and freshness. The fixed token value 1 represents one virtual query, not a secure multi-query protocol or an RF environment. These are explicit synthetic assumptions [15].

Average-energy begins with the coarse source integral and average power. Rich-static and complete-input supply all input records and the full model definition. They have the same available facts in this registered comparison, making their tie an admissible control. The saved condition name for complete-input is `complete-raw`; it does not mean that the participant receives the full coupled physical trace. That trace is host-side audit material. Executable-response adds candidate-specific public energy ledgers, activity and packets. Live E5 begins coarse and actually acquires permitted records, proposes a field or relation, obtains external qualification and submits a typed consumer policy. No target expression, required-field list or answer-bearing relevance label is prefilled [10, 12].

E4 changes only the rich-static initial serialization, comparing JSON with a typed JSON-Pointer table. The table preserves nested containers, types and empty values. All twelve formal pairs have saved receipts of identical decoded initial packets and tools. Later source responses use common JSON in both arms. This is an initial-serialization comparison rather than a representation intervention across the whole interactive trajectory.

6.2 Repetitions, resource parity and cache exposure

Two pilot repetitions on volatile-reset precede the formal comparison. The registered one-time instability rule selects three formal repetitions, then freezes that count. Pilots remain separate development evidence. The full eight-condition matrix contains 112 unique groups and 140 actual episodes: 96 formal groups and 16 pilot groups. Parent method proposals account for additional episodes in two conditions. This paper displays six

interface conditions, covering 72 formal and 12 pilot groups. Rich-static is also E4 JSON, and complete-input is also the fixed-method cell; these shared cells are counted once [10, 11].

Within each compared task, the scientific laws, service evaluator, legal design domain, reported model alias, initial cache and total resource caps remain fixed. Information access varies as declared by the condition. Each group has 24 provider requests, 40 tool calls, 1,200 s and at most two distinct design exposures, counting checks, response queries and confirmation. Acquisition, qualification and consumption do not add a free E5 budget. A parent proposal, when present, has an eight-request, 400 s sublimit within the group. The campaign caps are 3,000 requests, USD 200 in settled conservative estimates and 134,400 summed episode seconds. Failed groups are not automatically retried or replaced.

Every group starts from one immutable cache containing twelve exact-input responses and one registered derivative baseline. Private overlays isolate new job histories. A cache hit still consumes logical design exposure; it does not reveal another group's output before a query or enlarge the two-design cap. The records' term "physical cache" denotes the underlying digital execution cache and supplies no laboratory evidence. Fresh participant entries and the common cache identity are retained per group. The model is reported as `deepseek-flash`; this alias does not establish a pinned weight identity [11].

6.3 Preparation and negative controls

The structural preparation separately exports 64 inputs across four cases, four information levels, two formats and two observation regimes. Thirty-two format pairs preserve decoded facts, assets, tools and resource limits. These are preparation checks, distinct from actual model trials. Fixed parser checks abstain on all four current-only cases and correctly judge all four sufficient-information cases in both supplied serializations. They do not measure language-model comprehension [4, 6].

Negative controls retain the enriched static solution, complete raw model, uninformative observation, exhausted acquisition budget, illegal design changes and tampered source binding. A researcher-specified required-field selector and a researcher-supplied headroom expression are construction baselines, not autonomous discoveries. Preparation outcomes are not substituted for later live completion, and software validation of a consumer edge is not promoted to hardware acceptance.

7 Results: complete static information remains a strong control

7.1 The complete-service comparison

Table 3 reports every formal group. Average-energy yields 8 PASS and 4 FAIL. Rich-static, complete-input, executable-response and E4 table each yield 12 PASS. Live E5 yields 7 PASS and 5 INCOMPLETE; it has no admitted final service FAIL. Its completion rate is 7/12, not a conditional 7/7. Missing final executions and submitted service failures remain different outcomes with the same full denominator.

Table 3: Formal matched interface comparison: four fixed tasks, three repetitions per task, twelve groups per arm. P/F/I are PASS, submitted FAIL and INCOMPLETE. Every incomplete group and its requests remain included.

Condition	P	F	I	Requests	Requests/group
Average energy	8	4	0	75	6.2500
Rich static / E4 JSON	12	0	0	65	5.4167
Complete input	12	0	0	64	5.3333
Executable response	12	0	0	66	5.5000
Live observation (E5)	7	0	5	80	6.6667
E4 typed table	12	0	0	65	5.4167

The coarse arm's observed deficit occurs in the two reset tasks, each with one PASS and two FAIL. Pulse-fast and the C-lite derivative each have three PASS. Every strong arm passes all three repetitions on each task (Table 4). This finite result is consistent with the value of supplied timing and state information beyond the

Table 4: Formal P/F/I out of three per task. Each strong arm separately denotes rich static, complete input, executable response and E4 table; none has an omitted failure.

Fixed task	Average energy	Live E5	Each strong arm
Pulse-fast	3/0/0	2/0/1	3/0/0
Volatile reset	1/2/0	2/0/1	3/0/0
Latched reset	1/2/0	1/0/2	3/0/0
C-lite query derivative	3/0/0	2/0/1	3/0/0

coarse energy summary. It establishes no service advantage for executable-response over rich-static information. The saturated controls also do not prove equivalence on harder or previously unused tasks.

The two pilot repetitions are not pooled with the formal twelve. On their single volatile-reset input, average-energy has one PASS and one FAIL; each other displayed arm has two PASS. E5's two pilot completions include one selection and one construction. These are the development observations preceding the one-time formal-repeat freeze, not additional formal successes or independent tasks [12, 10].

7.2 What the seven prospective E5 chains did

Every formal E5 completion contains an actual prospective proposal, external source qualification and a typed observation consumed by the design policy. Six select existing fields and one constructs a relation. Five use component records and two excitation records. Six change the initial design and one does not. All seven final designs pass independent service, and all seven deliver before any reset. Their successes therefore establish neither pending-state survival across a reset nor the necessity of the acquired observation. New physical measurements number zero [12, 11].

Table 5: Disposition of all twelve formal E5 groups. Repeated rejected actions are not additional groups; absent executions are retained as INCOMPLETE.

Disposition	Groups	Final service
Qualified selection, consumed	6	PASS
Qualified construction, consumed	1	PASS
Source-unit binding rejected	4	INCOMPLETE
No typed consumer match	1	INCOMPLETE

Four incomplete groups retain `SOURCE_UNIT_UNAVAILABLE` in their original gateway records. The code identifies a submitted source/field/domain unit-binding problem; it does not imply that a physical quantity does not exist. In the pulse-fast example, the selected packet field `/packets/0/application_bytes` is unit-mapped in the acquired saved-activity source, while the declared domain also references deadline, TX-duration and readiness fields absent from that source's unit map. A latched-reset proposal selects a packet field from the model-law source, whose mapped fields describe model identity and Python definition. The original generic error does not name the first rejected term. We diagnose the source-local mismatch from saved proposals and logs; no repaired proposal is evaluated [13].

The fifth incomplete group records `NO_TYPED_CONSUMER_MATCH` on the C-lite derivative. Its equality expression outputs a Boolean while the consumer branches match numeric durations 0.002 and 0.001. A value can therefore be syntactically meaningful yet fail to enter the intended design edge. Other syntax and valid-proposal-cap errors remain charged actions inside these groups, not extra independent trials or free rescue attempts. Table 5 counts groups once.

The rejected original E5 attempt and its separately authorized post-hoc adapter chain remain outside these twelve formal groups. The post-hoc record demonstrates engineering linkage of an already generated relation. The seven later formal completions demonstrate prospective operation under a different registered protocol. Relabeling the former as a successful live discovery would erase both its original admission failure and the host intervention [7, 8].

7.3 Equal information and acquisition overhead

E4 gives identical complete-service and request totals: JSON and the typed table each pass 12/12 using 65 provider requests, or 5.4167 per group. The retained USD and wall receipts differ (Table 6), but this small comparison isolates neither all latency determinants nor the representation of later acquisitions. It supplies no observed service or request advantage and no causal format-speed or format-cost conclusion [12].

Table 6: E4 equal-information initial serialization. Later acquisitions use common JSON. Cost and summed group wall time are descriptive; neither establishes a format advantage.

Initial packet	PASS	Requests	USD	Wall (s)
JSON	12/12	65	0.4267200	676.5
Typed table	12/12	65	0.5145681	611.4

Table 7: Finite observation selection on four exposed cases. Read and audit bytes are Python payloads, not disk I/O. Host timings are single historical measurements, not latency benchmarks; selected and mechanism-spec contain the same facts.

Observation	Correct	Read B	Audit B	Wall ms	Model runs
uninformative	0/4	4648	4648	0.289	0
mechanism-spec	4/4	103	103	0.260	0
fixed-rich	4/4	4747	4747	0.282	0
full-raw	4/4	8544	8544	0.333	0
executable-dynamic	4/4	8544	8544	1.886	4
selected	4/4	103	103	0.236	0

The earlier finite observation selector chooses the smallest eligible source using researcher-supplied required-field metadata, without service outcomes. It reads the same 103 bytes and facts as the fixed mechanism-spec baseline; both correctly judge four cases. Rich-static, complete raw and an executable response also correctly judge all four. The historical timings in Table 7 include acquisition and source audit, plus selection when used. They are single-host payload measurements with uncontrolled cache state, not storage-I/O or monetary measurements. Thus lower payload volume relative to a full model is not an independent selection-algorithm gain [4].

A deterministic compiler also constructs the researcher-specified headroom observation. Its saved outputs of -14 and 8 μJ agree with twenty exposed clamp-grid checks and two fresh-process executions. Construction took 0.239 ms, reference qualification 13.017 ms and the two fresh processes 37.730 ms combined in those retained receipts. Because the equation and relevant fields were supplied by a researcher, this is executable construction machinery, not an autonomous discovery.

7.4 Service success versus an ordered retention witness

Four formal participant groups select a pending-event-across-reset result: two complete-input groups, one proposed-method group and one E4 table group. They make four fresh choices but reuse one distinct historical digital execution key. All select 10 μF and earliest TX at 45 ms. The same event is captured from 3.7 to 7.9 ms, a brownout occurs at 12.1 ms, and first valid reception follows at 46.9 ms. The fixed ideal passive latch provides the retention mechanism [14].

All twelve rich-static, all twelve executable-response and all seven successful E5 groups instead deliver before any reset. Their complete service passes are valid for their tasks but are not evidence for the stronger ordered-retention property. Conversely, four selections of one response are not four independent physical experiments or four newly invented retention mechanisms. The ordered audit creates zero new scientific executions and retains physical qualification UNKNOWN.

8 Event-time service exposes another interface boundary

8.1 The original O service and its strong fixed control

The O study extends the digital apparatus to vibration start, active keepalive and stop service. O0 is a 21,600 s task with zero initial capacitor energy, $C = 15$ mF and a prescribed source associated with a 15 cm^2 photovoltaic area. Its six intervals have nominal light levels 200/0/50/200/0/50 lux at boundaries 0/120/150/180/300/330/360 min. The corresponding electrical inputs are separately prescribed as 90/0/20/90/0/20 μW; the 3500 K label is metadata and lux is not converted into watts. RMS vibration is 0.08 g in the four declared activity intervals and 0.005 g otherwise. These inputs are uncalibrated digital assumptions [17].

Start requires RMS continuously above 0.05 g for 3 s; stop requires RMS continuously below 0.01 g for 60 s. Equality resets confirmation. The confirmation instant is the event time. Each start and stop must arrive within 10 s and carry an encoded timestamp within 1 s; active keepalive gaps must not exceed 120 s and idle false events fail. The first dark interval contains one required active interval, so darkness cannot simply be excluded from service.

The inherited ideal energy path has 0.8 harvest efficiency, 0.38 V/ 3 μW cold-start gates, 2.7/2.3 V ready/off thresholds, 3.3 V maximum capacitor voltage, 5.5 V rating and 0.3 Ω ESR. It supplies 1.8 V with 0.3 V dropout, 1/5 μW off/sleep power, 100 μJ boot and 130 μJ atomic TX transactions. Added RMS costs are 0.5 μW standby, 1 μJ wake, 20 μJ per successful sample and 5 μJ per committed write. The write is not charged at every poll. These proposed costs do not qualify a vibration sensor, PMIC, clock, NVM or RF receiver.

O0 compares a free legal candidate, a fixed 60 s periodic snapshot control and a fixed rich-state start/stop machine. They share the same task, source, hardware and event evaluator. The candidate and rich-state control submit the same one-second, NVM, `confirmed_events` program with the same thresholds, one-second retry and zero backoff. They share an exact execution key and response, match eight events, deliver 18 accepted packets and pass the nominal O predicate. The periodic control delivers 348 accepted packets and passes energy, but matches no required event. It fails service [16].

Table 8: Original O0 development task. Candidate and rich-state control share one exact design and response. Self-judgment is scoped separately from the host O event predicate.

Arm	Poll (s)	Matched/8	Host O	Self-judgment	Requests
candidate	1	8/8	PASS	UNKNOWN	8
fixed-periodic	60	0/8	FAIL	FAIL	5
rich-state	1	8/8	PASS	PASS	5

The candidate checks one-second sampling, then 1.5-second sampling, and ultimately submits the former. In the retained 1.5-second alternative, the first start is observed at 4204.070423 s and stamped 4204.139063 s against truth 4203 s. The stamp error 1.139063 s exceeds tolerance even though energy and delivery can appear adequate. This checked alternative is preserved in the same episode's exposure and cost history, not treated as a later repair to its accepted design [16].

The candidate's own judgment is UNKNOWN. Its original explanation concerns unobserved dark intervals after brownout, absent gap-report messages, inherited A/R first-report and dark-report metadata, and unavailable source/hardware calibration. In particular, inherited metadata calls for reception by 1800 s and dark reporting every 600 s, while the first confirmed O event is later. The host's O predicate asks a narrower event-service question and returns PASS. Its saved disagreement flag is nominal to that predicate. This is not evidence of a mistaken physical-acceptance judgment. Neither the flag nor O PASS discharges inherited reporting obligations. The observed brownouts near 8766.18 and 19587.64 s occur after required active intervals; success on this waveform does not establish uninterrupted six-hour sensing [16, 17].

8.2 A frozen new waveform and complete denominators

A secondary closed comparison freezes six descendant design procedures before any final input. Fixed H0 is unchanged; G0 is a fixed procedure improver; G1 is one actually proposed improver. Three paired descendant lineages are conditional on that single G1 formation, not three independent G1 training runs. Thirty effective selection cells pass before freezing. Each final design episode then sees the complete current task, including its waveform. "Unexposed" describes formation and selection, not concealment of the waveform from the final designer [19].

Each arm has nine final cells: three lineage contexts on each of two reserved digital tasks and a new O waveform. Both digital tasks pass in all eighteen cells. The complete 27-cell totals are H0 7/1/1, G0 6/1/2 and G1 8/0/1 in PASS/submitted-FAIL/INCOMPLETE order. The O rows account for all differences (Tables 9 and 10). G1's 8/9 is descriptive conditional evidence, not a stable improvement claim or a reason to replace H0 [18, 21].

Table 9: Complete 27-cell transfer denominator. Both reserved digital tasks pass in all 18 cells; O contributes the submitted failures and four incomplete episodes.

Procedure arm	PASS	Submitted FAIL	INCOMPLETE
H0	7	1	1
G0	6	1	2
G1	8	0	1

Table 10: All nine new-O final cells. Every submitted row matches eight events, receives 18 packets and passes energy. Events P/8 counts individually passing events. Dashes denote no submission.

Lineage/arm	Poll (s)	Retry (s)	Events P/8	Max error (s)	Service
L1-H0	1	1	6/8	1.103	FAIL
L1-G0	–	–	–	–	INCOMPLETE
L1-G1	0.75	0.25	8/8	0.728	PASS
L2-H0	0.75	5	8/8	0.728	PASS
L2-G0	–	–	–	–	INCOMPLETE
L2-G1	–	–	–	–	INCOMPLETE
L3-H0	–	–	–	–	INCOMPLETE
L3-G0	1	0.25	6/8	1.103	FAIL
L3-G1	0.75	1	8/8	0.728	PASS

All five submitted new-O designs pass energy, match all eight events and receive 18 accepted packets. Three designs sampled at 0.75 s pass complete service, with maximum encoded-time error 0.728 s. The two one-second submissions each pass six individual event checks and fail two stops with `WRONG_EVENT_TIME`. Their retry values differ, and their execution keys are distinct.

Table 11: The same two failed stop times occur in both one-second submissions. Error uses encoded event time, not receive time. Their distinct execution keys and different retry values remain preserved.

Confirmation (s)	Reception (s)	Encoded (s)	Error (s)
12077.625	12078.588	12078.629	1.004
17043.625	17044.588	17044.728	1.103

Both failed stop receptions occur less than one second after confirmation, but the encoded stamps are 1.004 and 1.103 s late (Table 11). Thus a correct match, prompt arrival, empty final queue and passing energy ledger cannot substitute for the encoded-time predicate. Detection and encoding must be checked together: for detection instant t_{det}, $|\hat{t} - t^*| \le |t_{\mathrm{det}} - t^*| + |\hat{t} - t_{\mathrm{det}}|$. The saved comparison does not isolate sampling phase from encoded clock behavior causally. Retry periods also vary, and one passing 0.75 s design belongs to H0. This is an observed design association, not a single-factor sampling ablation or an exclusive G1 mechanism.

Four O episodes have no submission: L1-G0, L2-G0, L2-G1 and L3-H0. Each last SDK assistant response reports stop reason `length`, 32,768 output tokens and 32,768 reasoning tokens, with thinking content and no final design or judgment. These are last-response counters, not the whole episode's token total or two additive token budgets. The outcomes remain INCOMPLETE; they are neither hardware failures nor energy FAILs. All requests, costs and nine O slots remain in the denominator, with no automatic retry [18].

The two submitted failures self-report PASS and therefore disagree with the nominal event predicate within the submitted scope. This differs from O0's broader UNKNOWN judgment. Missing submissions have no self-judgment and cannot be counted as zero-error successes.

9 Discussion and limitations

9.1 What changes when information is sufficient

The structural witnesses establish a precise failure of the current representation: identical present behavior can coexist with distinct responses to the same legal change. The enriched static controls show that the failure is not a universal static-versus-dynamic divide. Once the relevant law and state transition are supplied, the ideal tasks admit static prospective judgments. The matched comparison then tests actual finite-budget use, where the complete static arms already saturate service.

Live acquisition introduces more than information content. Source identity, source-local unit maps, domains, typed policy branches, proposal caps and final submission all become operational gates. The seven prospective E5 completions establish that this route can reach an actual consumer. Its five incomplete groups expose the cost of failing before execution. A compiler accepting a relation, an independent scalar agreement, a changed design and complete service are useful but different observations. None is automatically evidence of a necessary new fact or benefit over a sufficient static baseline.

The O result further narrows what the consumer must check. A representation containing only matched counts and energy PASS would omit the distinction that separates the two one-second failures from complete event service. Adding receive delay alone would still miss the encoded-time failure. The relevant interface must expose the predicate's actual operands. This is a concrete consumer-specific information requirement, not a blanket request for a larger trace.

9.2 Limits on generalization and acceptance

The four main tasks are fixed public development inputs with three formal repetitions, not a random sample of task families. Pilot evidence and formal evidence remain separate. The same reported model alias is used, but pinned weight identity is UNKNOWN. Source and evaluator hashes support execution identity; they cannot recover unavailable weight identity or establish an invoice audit. The current study does not estimate statistical significance, format equivalence, general interface superiority or stable recursive improvement.

The ideal clamp and memory models share assumptions across their independent implementations. The transient source is prescribed, the passive latch is ideal, and the C-lite source has no measured RF conversion or load response. O adds proposed RMS costs and a modeled clock, not measured sensor, PMIC, NVM or PHY behavior. Range consistency, conservation checks, software correctness and physical qualification therefore remain distinct. A digital PASS does not qualify source spectra, startup transients, leakage, clock uncertainty, write persistence or radio reliability.

Original A/R dark-report and strict FIFO failures remain preserved, with product status `NOT_ACCEPTED`. O event success neither erases those obligations nor accepts a complete product [22]. Likewise, avoiding a reset is not retaining state through one, and matching an event is not encoding its time correctly. These scope distinctions prevent an attractive sub-result from answering a broader untested question.

9.3 Accounting and retained host interventions

The full matched campaign records 771 SETTLED requests, USD 5.4478605 in conservative rate estimates and 8914.281914 summed episode seconds. It includes the two method-proposal conditions outside the six displayed

interface rows. E4 JSON is already the rich-static cell and is never added again. The secondary O/procedure campaign records 77 actual attempts for 73 logical cells, 367 SETTLED requests, USD 3.1767228 and 5951.639484 summed episode seconds. Its original O0 cost is a subset, not an additional campaign [11, 19, 21].

Four original selection host incidents are retained: three raise a missing historical-trace key and a fourth is censored by the hold. A registered continuation reconstructs the already sealed historical trace and links four named recovery attempts, preserving the original failures and valid earlier results. These infrastructure incidents are not negative G0/G1 scientific outcomes. Their 21 requests and USD 0.1318473 remain included in the 367-request total; they must not be added twice or silently removed [20].

All reported dollars are settled fixed-rate or conservative provider estimates, not invoice amounts. Historical UNKNOWN billing remains UNKNOWN in the coordinator's separate ledger. Codex assistance, historical acquisition and full end-to-end research cost are not priced by these episode sums. No hardware build or acceptance cost is inferred from them. Preparing this paper adds zero provider requests, scientific executions or native design mutations.

10 Saved-data reproducibility and implementation handoff

The accompanying `E_reproduce.py` uses only the Python standard library. Given `-data-root data` and `-output generated`, it verifies the portable input manifest and rebuilds E-series CSVs, all eleven numerical tables, counts, timestamp differences and preservation checks. It reads only package-relative saved files. Absolute paths inside historical records are archival strings and are never opened as fallback inputs. Missing or tampered data fails the check rather than dropping a group. No archived scientific Python is imported, and no simulator, observation qualifier, consumer or evaluator is rerun.

The package includes a data dictionary, claim-to-evidence index, case/observation implementation handoff and a model/version, exposure, intervention and cost index. Original syntax failures, post-hoc labels, incomplete groups, old UNKNOWN records' accounting boundary and product failures remain explicit. The shared official HKUST(GZ) style and vector mark are copied unchanged. Installed TeX and Poppler build and render the standalone manuscript; the document receipt records two-pass compilation, embedded fonts and actual per-page visual inspection. Saved-data reconstruction and document validation do not constitute a new scientific replication, fresh hardware test or external submission review.

11 Conclusion

An interface supports a legal design change when it distinguishes the world differences relevant to that consumer's prospective verdict. Current success need not provide that information, while complete static laws can. In the matched digital comparison, strong static and executable inputs each pass all twelve formal groups; prospective observation completes seven and leaves five incomplete. Actual typed consumption is demonstrated, but an advantage over complete static information is not. The event-time study shows a further boundary: correct matching and energy can coexist with wrong encoded timestamps. The resulting evidence supports explicit information, admission and service predicates, with failures, costs and UNKNOWN retained. Physical qualification remains UNKNOWN, the original product remains NOT_ACCEPTED, and default H0 remains unchanged.

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